

Research Statement

CIFRE PhD — CREST, Institut de Mathématiques de Toulouse, Generali France

*Regime-Conditional Factor Dependence, Supervised Capital Proxy,
and Deep Hedging for Insurance Total Portfolio Approach*

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1. INDUSTRIAL CONTEXT AND MOTIVATION

1.1. Starting point : 2022, a dependence crisis

The year 2022 was a major stress test for insurance asset managers. Unlike previous crises — which were essentially isolated shocks on a single asset class — 2022 saw equities and bonds fall simultaneously, inter-asset correlations turn positive, and traditional asset-class diversification break down entirely. For a life insurer such as Generali France, this episode exposed a fundamental structural flaw : Strategic Asset Allocation (SAA) models assume stable, linear dependence structures between economic factors, whereas the empirical evidence shows that these dependencies reverse abruptly in stress regimes.

This observation directly motivates the object of this thesis : *how to manage an insurance balance sheet when factor dependence becomes unstable, non-linear, and regime-dependent ?*

1.2. The current SAA process and its limitations

How the SAA process works today. The current SAA process at Generali France operates in three sequential steps, each carrying structural limitations.

Step 1 — Economic Scenario Generator (ESG). A proprietary ESG produces thousands of economic trajectories (interest rates, credit spreads, equity, real estate, inflation) over a long horizon, calibrated on historical data with **fixed correlations averaged across all regimes**. These trajectories feed all downstream actuarial calculations.

Step 2 — SCR calculation via nested LSMC. The Solvency Capital Requirement (SCR) is the regulatory capital under Solvency II, defined as the 99.5% one-year Value-at-Risk of net asset value change. Its calculation requires evaluating the Best Estimate of Liabilities (BEL) — the risk-neutral present value of all future policyholder cash flows — under each of the 10,000+ real-world scenarios produced by the ESG. Since each BEL evaluation requires a nested risk-neutral simulation (typically 1,000 inner paths), the raw computational cost is $10,000 \times 1,000 = 10$ million simulations. The industry-standard solution — Least-Squares Monte Carlo (LSMC, Krah et al., 2018) — replaces the inner simulation by a polynomial proxy $\hat{V}^L(\mathbf{x}) = \sum_k \beta_k \phi_k(\mathbf{x})$ fitted offline on a design-of-experiment. Even so, running Prophet / MoSes for a full SCR calculation takes **several hours**, making dynamic optimisation loops computationally infeasible.

Step 3 — Annual Markowitz optimisation and manual execution. The SCR is treated as a fixed constraint in a mean-variance optimisation over asset classes, revised **once per year**. Rebalancing instructions are transmitted **manually by email or Excel spreadsheet to GenAM** (Generali

Asset Management, the delegated manager), which then executes trades over several days. In addition, the optimisation criterion is fixed *ex ante* : there is no automatic identification of the dominant portfolio objective — for example duration control, inflation protection, carry harvesting, or capital efficiency — even though the relevant risk factor varies across portfolios and across regimes.

Four structural limitations.

1. **Regime-blind dependence.** The ESG calibrates correlations on long historical averages. In 2022, the equity/bond correlation switched from -0.3 (diversifying) to $+0.6$ (concentration risk) within weeks. A model with fixed correlations cannot anticipate such reversals : the Markowitz optimum for a low-inflation regime is a high-risk portfolio in a stagflation regime.
2. **LSMC proxy limitations.** The polynomial LSMC proxy is accurate only within the region covered by the design-of-experiment, which is constructed without regime conditioning. Precision degrades precisely in the stress zones most relevant for capital decisions. Moreover, the proxy is not expressed in terms of TPA factors, preventing its direct integration into a factor-based allocation loop.
3. **Static, siloed decision-making.** Markowitz optimises a single-period mean-variance criterion without anticipating future regime transitions or internalising hedging costs. Allocation and hedging are decided separately, yielding a suboptimal joint policy for an insurer whose critical losses are convex. Moreover, the objective function is imposed uniformly rather than identified automatically from the portfolio's dominant risk factor and balance-sheet context, which is precisely what a dynamic TPA process should adapt over time.
4. **Private asset valuation.** Private equity, private debt, infrastructure and real estate constitute a growing share of Generali's portfolio. Their quarterly, smoothed NAV valuations create an **illusion of low correlation** with public markets : in 2008, private equity NAVs remained stable for 6–12 months after the crash before correcting sharply. This artificial smoothing *understates* true economic dependence in stress regimes and introduces a lag bias in the SCR calculation. Furthermore, the irrevocable capital commitments of closed-end funds are incompatible with dynamic rebalancing frameworks that treat all assets as freely tradeable.

1.3. The TPA as target framework

The **Total Portfolio Approach (TPA)** is the management framework that Generali France aims to adopt, evaluating each investment decision through its marginal contribution to total balance sheet risk via common economic factors : Duration, Inflation, Growth, Liquidity, Carry. Implementations at sovereign wealth funds (Future Fund, CPP Investments) report approximately $+1\%$ excess return per year relative to classical SAA (Thinking Ahead Institute, 2025 ; CAIA, 2024).

However, TPA currently rests on simplistic dependence assumptions : linear, Gaussian, non-regime-conditional. For an insurance company, the factor space must also include liability-side and cross-cutting factors (surrender rate dynamics, longevity, yield curve interactions with the BEL). This thesis provides the missing quantitative foundations to make TPA operational for an insurer under Solvency II, structured along three research axes : regime-conditional factor simulation (Axis 1), fast capital proxy (Axis 2), and joint allocation-hedging policy (Axis 3).

2. RESEARCH QUESTIONS

Research Question 1 (Regime-conditional factor dependence : from GAN, VAE and SGM to formal guarantees). *How to simulate the joint distribution of TPA factors conditionally on the current macroeconomic regime — identified in real time by a sequential filter — while capturing the non-linear, asymmetric tail dependencies relevant for extreme risk ? Different generative*

methodologies will be tested and compared : Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs) and Score-Based Generative Models (SGMs). Specifically : (i) how to formally validate the conditional dependence structure learned by a regime-conditional GAN or VAE using copula specification tests ; and (ii) how to obtain W_2 -convergence guarantees when upgrading to an SGM with sequentially estimated regime states ?

Research Question 2 (Supervised SCR proxy for dynamic decision-making). *How to construct a proxy for the insurance SCR — expressed as a function of TPA factors, the current regime, and key liability states including private asset NAV lags — that is sufficiently accurate in decision-relevant zones (near the regulatory threshold, under stress regimes), with controlled error bounds per regime and sub-10ms inference time, enabling its integration into a dynamic optimisation loop as an explicit capital constraint in the allocation programme ?*

Research Question 3 (Joint allocation and deep hedging policy). *How to learn a dynamic decision policy jointly optimising TPA factor exposures and explicit hedging positions (deep hedging), with a portfolio-specific objective function automatically inferred from the dominant risk factor to optimise, and under capital and liquidity constraints including an explicit SCR proxy constraint and private asset engagement limits, in the presence of unstable, regime-conditional factor dependencies, so as to reduce the tail risk of the insurance balance sheet ?*

3. RELATED WORK AND POSITIONING

3.1. Generative models and regime-conditional dependence

Generative Adversarial Networks applied to financial series have demonstrated their capacity to capture multivariate stylized facts : heavy tails, volatility clustering, and asymmetries (Wiese et al., 2020, Quant-GAN ; Yoon et al., 2019, TimeGAN). Conditional GAN architectures (cGAN) allow generation to be conditioned on external variables — here, the macroeconomic regime. However, GANs suffer from *mode collapse* and provide no formal guarantees on the quality of the generated distribution, limiting their regulatory auditability.

Variational Autoencoders (VAEs) offer a complementary perspective : the data is encoded into a structured latent space $\mathbf{z} \sim q_\phi(\mathbf{z}|x)$ and the generative distribution $p_\theta(x|\mathbf{z})$ is learned by maximising the evidence lower bound (ELBO) :

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|x)}[\log p_\theta(x|\mathbf{z})] - \text{KL}(q_\phi(\mathbf{z}|x) \parallel p(\mathbf{z})).$$

Unlike GANs, VAEs provide a stable training objective and an explicit probabilistic framework, but can produce overly smooth samples due to the Gaussian posterior approximation. Conditional VAEs (CVAEs) allow regime-based conditioning via a label-conditioned prior $p(\mathbf{z}|k)$, providing an auditable latent representation of each regime’s dependence structure.

Copula theory (Sklar, 1959) provides a complementary framework for validating learned dependence. By Sklar’s theorem :

$$P(\mathbf{F}_t \leq \mathbf{f}) = C(F_1(f_1), \dots, F_d(f_d)),$$

where C is the copula capturing the dependence structure independently of the marginals. Fermanian (2013, 2022) has developed specification tests for conditional copulas, providing the statistical tools to formally validate that a GAN, VAE or SGM has correctly captured the conditional tail dependence coefficient λ_k per regime.

Score-Based Generative Models (SGMs) offer a generative alternative with formal convergence guarantees. Strasman, Ocello et al. (TMLR, 2024) establish explicit bounds on the Wasserstein-2 distance as a function of the noise schedule $g(t)$:

$$\text{KL}(\hat{p} \parallel p^*) \leq \int_0^T g(t)^2 \varepsilon_{\text{score}}(t) dt + \varepsilon_{\text{init}}. \quad (1)$$

Gentiloni-Silveri and Ocello (ICML, 2025) extend these guarantees to non-log-concave distributions — notably Gaussian mixtures corresponding to distinct market regimes.

At the frontier of generative modelling for time series with structural breaks, De Marco, Pham & Zanni (2025) introduce a jump-diffusion Schrödinger bridge model extending the diffusion-based approach to accommodate discontinuities in the generative dynamics. Starting from a Schrödinger bridge entropy-minimisation problem, the model characterises the optimal controlled jump-diffusion process generating synthetic time series matching the joint distribution of observed data — including abrupt movements, heavy tails and regime changes that diffusion-only models fail to reproduce.

Comparison metrics. A rigorous comparison of GAN, VAE and SGM for regime-conditional factor generation will be conducted along four axes : (i) *statistical fidelity* — Wasserstein-2 distance W_2 and Maximum Mean Discrepancy (MMD) between generated and empirical distributions ; (ii) *tail calibration* — per-regime conditional tail dependence λ_k^{gen} versus empirical estimates, validated via Fermanian’s (2013) specification tests ; (iii) *temporal structure* — reproduction of autocorrelation and cross-autocorrelation patterns, volatility clustering and leverage-effect signatures ; (iv) *regulatory auditability* — presence of formal convergence bounds (W_2 for SGM ; ELBO for VAE), interpretability of latent regime structure, and sensitivity to training sample size.

Identified gap. Neither conditional GANs, conditional VAEs, nor conditional SGMs have been applied to simulating the joint distribution of TPA factors with real-time regime identification. For the GAN and VAE : the absence of formal regime-conditional tail dependence validation via copula specification tests. For the SGM : the absence of W_2 bounds for the conditional case when the regime state is estimated sequentially with inference error. These contributions constitute Axis 1.

3.2. Insurance liability proxy models

The LSMC framework of Krah et al. (2018) is the industry reference for life insurance proxy models, approximating the BEL by polynomial regression on a pre-computed design-of-experiment. This approach presents two critical limitations : it does not condition the SCR on the current macroeconomic regime, and its accuracy degrades precisely in the stress zones most relevant for the allocation decision. Neural network approaches (Huber & Schmock, 2022) have demonstrated that a well-specified proxy can achieve 1–3% accuracy with inference times below 10ms.

From the perspective of stochastic approximation methods for financial risk measures, Bourgey (2020) provides a comprehensive analysis including multi-level Monte Carlo estimators for nested expectations — directly relevant to the LSMC nested SCR problem — and polynomial chaos expansion meta-models for large weighted sums of conditionally independent Bernoulli variables, with L^2 error estimates as a function of the expansion order. These results motivate the adaptive-weighting architecture of Axis 2, where accuracy is concentrated near the regulatory threshold and in stress regimes.

Identified gap. Existing proxies are constructed independently of the regime and TPA factors, and none accounts for the NAV-lag bias of private assets in the SCR calculation. A regime-conditioned, factorially coherent proxy with adaptive sample weighting and controlled error bounds does not exist in the literature. This is the contribution of RQ2.

3.3. Deep RL, dynamic allocation and deep hedging

Wekwete (2023) demonstrated the applicability of Deep RL (PPO, SAC) to insurance ALM in stochastic environments. Buehler et al. (2019) introduced *deep hedging* : learning an optimal hedging strategy via a neural network minimising a convex risk measure, without market completeness assumptions. This approach has been extended to extreme risk hedging (Carbonneau, 2021 ; Cao et al., 2023) and multi-asset portfolios under stress regimes.

On the risk-measure side, Rosazza Gianin et al. (2024) develop a theory of shortfall systemic

risk measures and capital allocation rules, providing axiomatic foundations and tractable capital allocation schemes for multi-entity portfolios under dependence via a generalised collapse-to-the-mean framework. Their results connect naturally to the distortion-based capital constraint embedded in Axis 3.

Réveillac (2013) studies optimal investment with CRRA preferences subject to dynamic risk constraints on trading strategies. Optimal policies are characterised by a **quadratic backward stochastic differential equation (BSDE)**, and within the class of time-consistent distortion risk measures a three-fund separation result is established. This framework directly informs the stochastic control formulation of Axis 3, where the joint allocation-hedging policy operates under a dynamic SCR constraint generated by a general class of risk measures.

Identified gap. Learning a joint allocation-and-hedging policy — fed by an unstable dependence model and a fast SCR proxy, respecting private asset engagement constraints, while automatically selecting the relevant portfolio objective in factor space and being validated through out-of-sample allocation backtests within a TPA insurance framework under Solvency II — does not exist in the literature. This is the contribution of RQ3.

4. METHODOLOGY AND DELIVERABLES

4.1. Axis 1 — Regime-conditional factor dependence (Years 1–2)

Axis 1 proceeds in two methodological stages corresponding to a progressive increase in theoretical rigour.

Stage 1 — Regime-conditional GAN (Year 1). Let $z_t \in \{1, \dots, K\}$ be the macroeconomic regime identified in real time by a particle filter (SMC, Chopin & Papaspiliopoulos, 2020). A conditional GAN is trained to simulate the joint distribution of TPA factors conditionally on the current regime :

$$\mathbf{F}_{t+1} \mid z_t = k \sim G_\theta(\varepsilon, k), \quad \varepsilon \sim \mathcal{N}(0, I), \quad (2)$$

where G_θ is the generator and the discriminator D_ϕ distinguishes real from synthetic scenarios regime by regime. The advantage of the GAN is total flexibility : no parametric assumption on the dependence structure is imposed, allowing the model to learn asymmetries and non-linearities in tail dependence under stress. The conditional tail dependence coefficient

$$\lambda_k = \lim_{u \rightarrow 1} P(F_i > F_{i,k}^{-1}(u) \mid F_j > F_{j,k}^{-1}(u), z_t = k)$$

is estimated empirically and validated via Fermanian’s (2013) specification tests for conditional copulas.

Private asset treatment. The factor vector includes a *vintage lag factor* $\tilde{F}_t^{\text{PA}} = \text{NAV}_{t-1}^{\text{PA}}$ capturing the quarterly smoothing bias of private equity, real estate and infrastructure NAVs. The GAN learns the true economic dependence between private and public assets, including the delayed correction dynamic observed in 2008 and 2020.

Stage 1b — Regime-conditional VAE. As a complementary generative approach within Stage 1, a **Conditional VAE (CVAE)** is trained alongside the GAN. The encoder maps each historical factor trajectory ($\mathbf{F}_{t+1}, z_t = k$) to a regime-conditional posterior $q_\phi(\mathbf{z} \mid \mathbf{F}_{t+1}, k)$, and the decoder generates new factor vectors from samples of the prior $p(\mathbf{z} \mid k)$. The ELBO provides a stable training objective free from mode collapse, at the cost of potentially smoother marginal distributions. Both GAN and VAE outputs are evaluated against the comparison metrics defined in Section 3.1, with particular attention to regime-conditional tail calibration.

Stage 2 — SGM with W_2 guarantees (Year 2). The GAN, while empirically powerful, suffers from mode collapse and provides no formal guarantees — limiting regulatory auditability. The second stage replaces or complements the GAN with a conditional SGM, building on the

theoretical framework of Strasman, Ocello et al. (TMLR, 2024) and Gentiloni-Silveri & Ocello (ICML, 2025). The conditional OU forward process for regime $z_t = k$ is :

$$d\mathbf{x}^\tau = -\mathbf{x}^\tau d\tau + \sqrt{2} g_k(\tau) dW_\tau, \quad \tau \in [0, T], \quad (3)$$

with regime-specific noise schedule $g_k(\tau)$. The convergence bounds give :

$$W_2(\hat{p}_k, p_k^*)^2 \leq C \int_0^T g_k(\tau)^2 \varepsilon_{\text{score}}(\tau) d\tau,$$

even for non-log-concave distributions corresponding to distinct market regimes.

Theoretical contribution. (i) Empirical validation of tail dependence λ_k generated by the GAN via Fermanian (2013) specification tests. (ii) Extension of the W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online HMM — the key open question : how do these bounds evolve when regime z_t is estimated sequentially with particle filter inference error ?

Scenario selection layer. Beyond generating a full scenario cloud, Axis 1 also includes a scenario selection step producing a **central regime-conditional scenario** used as the baseline path for reporting, comparison, and downstream optimisation diagnostics.

4.2. Axis 2 — Supervised SCR proxy (Year 2)

Model. Let $\mathbf{P}_t \in \mathbb{R}^p$ be the liability state vector : effective duration, average guaranteed rate, observed lapse ratio, and the private asset vintage lag $\tilde{F}_{t-1}^{\text{PA}}$ capturing NAV smoothing bias. We learn a supervised proxy :

$$\widehat{\text{SCR}}(t) = f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t), \quad (4)$$

where f_θ is a neural network (MLP + attention) trained on a design-of-experiment $\{(\mathbf{F}^{(i)}, z^{(i)}, \mathbf{P}^{(i)}), \text{SCR}^{(i)}\}$ computed offline via Prophet/MoSes. The key methodological innovation is **adaptive weighting** : scenarios close to the regulatory threshold and in stress regimes receive higher training weight, concentrating accuracy where the decision is most sensitive. The generalisation bound satisfies :

$$|\text{SCR}(t) - f_\theta(\mathbf{F}_t, z_t, \mathbf{P}_t)| \leq \varepsilon_k(n_k, \delta), \quad (5)$$

where $\varepsilon_k(n_k, \delta)$ decreases with the number of training points n_k in regime k and with confidence level $1 - \delta$.

Contribution. First SCR proxy jointly conditioned on the macroeconomic regime, TPA factors, and private asset NAV lag, with empirical error bounds validated across portfolio ranges including Fonds Euro and Euro-Croissance. Inference in < 5 ms versus several hours for the full Prophet/MoSes calculation. This proxy is not only a reporting object : it enters the downstream allocation programme as an explicit state-dependent capital constraint.

4.3. Axis 3 — Joint allocation and deep hedging policy (Year 3)

Model. For each portfolio, a latent objective map $\psi_t = g_\eta(\mathbf{F}_t, z_t, \mathbf{P}_t, \widehat{\text{SCR}}_t)$ identifies the dominant factor objective to optimise (for example duration control, inflation hedging, carry extraction, liquidity preservation, or capital efficiency) as a function of the current regime, liability state and capital position. This layer is what allows the framework to move from a static allocation rule to a genuinely dynamic allocation policy.

The agent state is $s_t = (\mathbf{F}_t, \hat{C}_{z_t}, z_t, \mathbf{P}_t, \widehat{\text{SCR}}_t)$, where $\hat{C}_{z_t}$ is the regime-estimated copula from Axis 1 and $\widehat{\text{SCR}}_t$ the proxy from Axis 2. The action is the pair $(\mathbf{w}_t, \mathbf{h}_t)$, where $\mathbf{w}_t \in \Delta^{d-1}$ is the TPA factor allocation vector and $\mathbf{h}_t \in \mathcal{H}$ is the hedging position (puts, overlays, convexity structures). The policy π maximises :

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \left(r_{\psi_t}(\mathbf{w}_t, \mathbf{h}_t, \mathbf{F}_{t+1}) - \lambda_1 c(\mathbf{h}_t) - \lambda_2 \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1 - \lambda_3 \mathbf{1}[\Delta w_t^{\text{PA}} > 0] \right) \right], \quad (6)$$

subject to the explicit capital constraint

$$\widehat{\text{SCR}}_t \leq \bar{C},$$

where r_{ψ_t} is the portfolio-specific objective functional selected in factor space, $\bar{C}$ the regulatory capital budget, $c(\mathbf{h}_t)$ the hedging cost (option premium, bid-ask spread), and the term $\lambda_3 \mathbf{1}[\Delta w_t^{\text{PA}} > 0]$ penalises new private asset commitments to reflect their irrevocable, illiquid nature.

Integrated deep hedging. Unlike classical deep hedging (Buehler et al., 2019) which targets payoff replication, the hedging component here aims to reduce the $\text{CVaR}_{0.99}$ of the capital deficit while internalising hedging costs. The regime-conditional factor dependence of Axis 1 is directly embedded in the simulation environment, forcing the policy to be robust to dependence regime switches — including the private asset delayed correction dynamic.

5. EXPECTED CONTRIBUTIONS

Type	Description
<i>Theoretical</i>	(i) Non-asymptotic error bounds on estimated conditional tail dependence λ_k for regime-conditional copulas and GANs (Fermanian); (ii) Extension of W_2 bounds of Gentiloni-Silveri & Ocello (ICML 2025) to the conditional SGM case with online SMC regime estimation (Ocello); (iii) Generalisation bounds for the regime-conditioned SCR proxy with adaptive weighting and private asset NAV lag.
<i>Methodological</i>	(i) HMM-SMC architecture coupled to a regime-conditional GAN/VAE/SGM with private asset vintage lag factor; (ii) Supervised SCR proxy conditioned on regime and TPA factors with adaptive weighting and explicit integration as optimisation constraint; (iii) Dual allocation-and-hedging agent under private asset engagement constraints and automatic portfolio-objective identification.
<i>Industrial</i>	Operational tools deployed at Generali : auditable regime-conditional scenario generation, near-instantaneous SCR computation as optimisation constraint, TPA allocation with integrated hedging and GenAM interface.
<i>Regulatory</i>	Models with controlled, auditable guarantees — W_2 bounds for the generative module, empirical error bounds for the SCR proxy — directly usable by the ACPR and the Generali Group Risk Division.

6. TIMELINE (36 MONTHS)

Phase	Period	Key deliverables
1	1st year	Bloomberg factor database (20 years) including private asset NAV lags ; online HMM-SMC calibration ; regime-conditional GAN/VAE with copula validation on historical crises ; paper submission (Journal of Multivariate Analysis or Finance & Stochastics).
2	2nd year	Actuarial design-of-experiment (Prophet/MoSeS) ; training and validation of regime-conditioned SCR proxy with private asset NAV lag ; integration of the SCR proxy as an explicit optimisation constraint ; empirical error bounds per regime and portfolio range ; paper submission (Insurance : Mathematics and Economics).
3	3rd year	Training of dual allocation-and-hedging agent with private asset engagement constraints and automatic objective selection per portfolio ; full three-axis coupling ; multi-regime out-of-sample backtesting (2008, 2020, 2022) ; PhD defence and publications (Quantitative Finance or JFQA).

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